

Townsend 1.1-1.3

Questions for Gallicchio

I don't understand what changes fundamentally between the SG experiments where we block the $|-Z\rangle$ particles, and recombine to see $|\pm Z\rangle$ vs when we don't block but they come in $|+Z\rangle$ so we only measure that.

1.1: Stern Gerlach

Lets see why spin angular momentum of a subatomic particle is different from orbital angular momentum.

Magnetic Dipole Moment

$$\mu = \frac{IA}{c} = \left(\frac{q}{T}\right) \frac{\pi r^2}{c} = \frac{qvr}{2c} = \frac{q}{2mc} L$$

where A is area of circle, I is current, q is charge, T is orbital period $T = \frac{2\pi r}{v}$ and $L = mvr$ is the orbital angular momentum $I = \frac{q}{T}$.

Orbital spin $\mu = \frac{q}{2mc} L$.

Intrinsic spin $\mu = \frac{gq}{2mc} S$, where g is a proportionality constant.

Stern Gerlach experiment: melted silver isn't polarized but is sent in a steady stream through a nonconstant magnetic field. We don't see a continuum of deflection like we would expect classically.

1.2: 4 experiments

If we have a stream of $|+z\rangle$ going through a z oriented Stern Gerlach, we would only see $|+z\rangle$. If we feed that pure $|+z\rangle$ to an x oriented SG, then we would see half $|+x\rangle$ and half $|-x\rangle$. Putting it through another SG in the z would give us... not pure $|+z\rangle$, but instead half each $|\pm z\rangle$.

S_z and S_x are incompatible observables.

1.3: State vector

$\langle -z | +z \rangle = 0$ means that the probability of measuring a particle in the down state if it was known to be up is zero.